

Supplementary Materials:

TITLE: Does Simplifying Search for Qualified Racial Minorities Boost STEM Diversity? A Field Experiment

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Data and Code Availability: All analysis code, data files, and an extended appendix containing additional tables reporting treatment effects on all Underrepresented Minority (URM) and Hispanic speaker outcomes are available on the Open Science Framework at <https://osf.io/adg4p/overview>.

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Section 1: Intent-to-Treat Robustness Test of Primary Analyses

We conducted a pre-registered intent-to treat (ITT) robustness test of our main analysis using the full randomized sample of 1,881 seminars (not just the 1,686 seminars that continued to hold events in the 2024-25 academic year and to list speakers online, allowing us to measure our pre-registered outcomes of interest). In these analyses, we assumed zero Black speakers appeared in missing seminars. Table S6 reports the regression-estimated effects of assignment to the URM Directory Treatment condition on Black speaker outcomes among the full randomized sample ($N=1,881$) using our primary pre-registered regression specifications with both extended controls (Models 2, 4, and 6) and with more limited controls (Models 1, 3, and 5). We find that assignment to treatment significantly increased the percentage of Black speakers per seminar (Table S6, Model 2, $B = 0.592$, $SE = 0.263$, $p = .025$), the number of Black speakers per seminar (Table S6, Model 4, $B = 0.066$, $SE = 0.033$, $p = .049$), and the likelihood that a seminar featured at least one Black speaker (Table S6, Model 6, $B = 0.044$, $SE = 0.020$, $p = .030$) in a seminar series. These results suggest that our main findings are robust to the inclusion of unobserved seminars under conservative assumptions about their outcomes.

We conducted analogous ITT analyses for Underrepresented Minority (URM) and Hispanic speaker outcomes. Consistent with the lack of robust main effects reported in the main text, treatment effects on URM and Hispanic speaker representation remained non-significant under the ITT specification across all outcome measures (all p 's $> .13$; see Extended Tables E15–E16 on OSF).

Section 2: Demographic Classification

2.1 Classification of Seminar Speakers

We classified the race and gender of seminar speakers using a two-pronged approach that combined facial image analysis with name-based inference. For each of the 17,712 unique speakers in our sample (appearing across 23,455 seminar-speaker slots), we first attempted to retrieve a photograph via automated Google Image searches using the speaker's name and institutional affiliation as search terms. Retrieved photographs were then analyzed using DeepFace (Parkhi et al., 2015), an open-source facial analysis framework that estimates demographic attributes—including race and gender—from facial features. In parallel, we generated name-based demographic predictions using OpenAI's GPT-4o model, which inferred each speaker's likely race and gender from their full name.

The GPT-4o model was queried at temperature 0.1 and prompted to respond in JSON format, providing a race and gender classification along with a confidence score for each. The exact prompt supplied to the model was as follows.

System message: You are an expert in demographic analysis based solely on names. You MUST always provide a definitive answer choosing from the allowed categories only. Never use 'unknown' or refuse to make a determination.

User message:

Analyze the demographic characteristics based solely on this name:

Name: {name}

Please provide your best assessment of:

1. Gender - Use ONLY these categories:

- Male
- Female

(You MUST choose one - no unknown/non-binary options)

2. Race/Ethnicity - Use ONLY these categories:

- white (includes European and Middle Eastern origins)
- black (African or African American origins)
- latino (Hispanic or Latino origins)
- asian (includes East Asian, South Asian, and Indian origins)
- native american (Native American origins)

(You MUST choose one - no unknown/mixed/other options)

IMPORTANT:

- You MUST make a determination even if uncertain - choose the most likely option
- If the name could belong to multiple categories, choose the single most likely one
- Use lower confidence scores (0.3-0.5) when uncertain, but still make a choice

Respond in JSON format:

```
{  
  "gender": "your_assessment",  
  "gender_confidence": 0.0-1.0,  
  "race": "your_assessment",  
  "race_confidence": 0.0-1.0,  
  "reasoning": "brief explanation"  
}
```

Both the face-based and name-based approaches produced categorical predictions for race and gender (male or female). The face-based model (DeepFace) natively outputs six race categories (White, Black, Asian, Indian, Middle Eastern, and Latino Hispanic). Our classification pipeline collapses Indian into Asian and Middle Eastern into White, so four DeepFace-derived categories enter the combined classification, with Native American contributed only by the name-based (GPT-4o) path. The name-based model (GPT-4o) classified race into five categories (White, Black, Hispanic/Latino, Asian, or Native American). Demographic labels were assigned using a hierarchical decision rule that prioritized face-based predictions when the facial analysis produced a confidence score exceeding 90% (this was the case for 38% of speakers in our sample); otherwise, we relied on the name-based prediction algorithm (this was the case for 62% of speakers in our sample). This approach reflects prior work showing that name-based race inference is inherently probabilistic and varies substantially in accuracy across racial groups (Imai & Khanna, 2016; Kozlowski et al., 2022), motivating the prioritization of face-based predictions when a high-confidence facial image is available, while ensuring broad coverage through name-based inference for remaining cases.

To confirm the validity of this algorithmic classification, a research assistant independently coded the race and gender of a random subset of 200 speakers and achieved 100% agreement with the automated classifications. Throughout the paper and this appendix, we use the term “Black” to refer to individuals classified as “Black non-Hispanic.”

2.2 Classification of Seminar Organizers and Department Chairs

We classified the race and gender of 567 department chairs and 1,019 seminar organizers using the same two-pronged approach described in Section 2.1. Where classification relied on the name-based path, recipients were classified with an abbreviated variant of the GPT-4o prompt described in Section 2.1, using the same race and gender category lists but without the reasoning field. The full text of this abbreviated prompt is available in the replication package. Face-based predictions met the 90% confidence threshold for 32% of department chairs and 35% of seminar organizers; the remainder were classified via name-based inference. Of the 1,019 seminar organizers, 49 (4.8%) could not have their race and gender classified because the seminar contact listed was a generic departmental email address (e.g., physics-seminar@university.edu) rather than a named, individual academic; these observations were excluded when computing department-level recipient demographic covariates. All 567 department chairs’ demographics were successfully classified.

2.3 Classification of Faculty Listed in the URM Directory

The 550 faculty members in the URM directories provided in our treatment condition had their demographics classified manually by research assistants as part of the directory construction process described in Appendix Section 3.1 (below). Each faculty member's race (Black, Hispanic, or Native American) and gender were recorded by research assistants based on information available on each faculty member's departmental profile, including photographs and

biographical details. To validate these classifications, an independent research assistant who was not involved in the original directory construction coded the race and gender of a random sample of 100 faculty members who were included in the original URM Directory. The research assistant was shown each faculty member's photograph, name, title, university and website, and encouraged to search for any additional information that could help them verify the faculty member's demographic identity. Agreement between the independent RA's coding and the original directory classifications was 92% for race and 98% for gender.

Section 3: Directory Construction

As described in the main manuscript, seminar organizers and department chairs from each department in the treatment and control conditions were provided with access to a discipline-specific directory of underrepresented minority (URM) faculty or a list of peer department homepages, respectively. The directories were constructed as follows.

3.1 URM Directory (Treatment Condition)

Departments assigned to the *URM Directory Treatment* condition received access to a curated, discipline-specific directory of URM faculty at peer institutions (see Figure S3 for an example). Peer departments were defined as those ranked within plus or minus 20 positions of the host department in the 2024 U.S. News & World Report discipline-specific rankings, with the peer set iteratively expanded one upward and then downward until at least 20 URM faculty at peer departments were identified for inclusion. A minimum of 40 peer departments was ensured by expanding the set of universities near the top or bottom of the ranking, so the count of peer departments used as a control variable occasionally exceeds 40.

Five discipline-specific lists of URM faculty were constructed from which the set of 20+ peer faculty were drawn for each personalized directory shared with a given department chair and seminar organizer. One list was created for each STEM field in the sample: Chemistry (a list of 108 URM faculty), Computer Science (a list of 94 URM faculty), Mathematics (a list of 105 URM faculty), Mechanical Engineering (a list of 131 URM faculty), and Physics (a list of 112 URM faculty), yielding 550 URM faculty entries in total across all lists. Of all the faculty included on these lists, 268 (48.7%) were classified as Black, 280 (50.9%) as Hispanic, and 2 (0.4%) as Native American; 360 (65.5%) were men and 190 (34.5%) were women. The average

directory listing URM faculty from peer departments that we sent to a seminar organizer and department chair included 31 URM faculty (min = 20, max = 64).

To construct the lists of URM faculty that populated the personalized directories shared with those in our treatment condition, three research assistants systematically visited the department websites of all 146 R1 universities in our sample and identified faculty who were underrepresented racial minorities—defined as Black, Hispanic, or Native American—by reviewing faculty profile pages and photographs. Each directory entry included a photograph, name, academic title, institutional affiliation, race, gender, research area(s), and a link to the faculty member’s webpage (see Figure S3 for an example screenshot). To assess the reliability of these classifications, an independent research assistant who was not involved in the original directory construction coded the race and gender of a random sample of 100 faculty members using each faculty member’s photograph, name, title, university, and website. Agreement between the independent research assistant’s coding and the original directory classifications was 92% for race and 100% for gender.

3.2 Control Directory

Departments assigned to the *Directory Control* condition received a directory listing peer department names with hyperlinks to department homepages (see Figure S4 for an example). Peer departments were defined using the very same U.S. News and World Report ranking-based criteria as in the treatment condition.

Section 4: Speaker-Directory Matching Protocol

We assessed whether each of the people who filled one of the 23,455 STEM seminar-speaker slots under study in the 2024-25 academic year appeared in (or, in the control condition, *would* have appeared in) the relevant URM directory for that seminar's organizer.

Because names appear in varied forms across data sources—differing in punctuation, middle initials, suffixes, and occasional misspellings—we employed a four-step matching protocol applied identically to Black and Hispanic speakers. First, all speaker and directory names were normalized by converting to lowercase, removing all non-alphabetic characters except spaces, collapsing consecutive whitespace, and trimming. Second, normalized speaker names were compared to normalized directory names via exact string equality. This “exact matching” process confirmed that 33 Black and 29 Hispanic seminar speaker appearances were filled by someone who appeared in a seminar organizer's relevant directory. Third, for unmatched speakers, we computed the Jaro-Winkler distance (Jaro, 1989; Winkler, 1990) between every remaining speaker–directory name pair and flagged all pairs with a distance of 0.20 or below as candidates for manual review. Fourth, each candidate pair with a Jaro-Winkler distance of 0.20 or below was manually reviewed by two independent research assistants blind to condition who verified whether the two names plausibly referred to the same individual. Combining exact and fuzzy matching, this procedure identified a total of 41 unique Black and 46 unique Hispanic individuals who appeared in both a seminar lineup and a relevant organizer's directory.

Section 5: Supplementary Figures

Figure S1. Sample Recruitment Email

From: Seminar Support (seminar-support@wharton.upenn.edu)

Sent:

To:

Subject: your seminar series

Dear [Professor X/Seminar Organizer],

We're a non-profit at the University of Pennsylvania contacting you in your capacity as the [current chair of your department/organizer of your {seminar}].

To support seminar organizers, we have created a database of [scholars in your field who might/ departments in your field that you can search to] enhance the diversity of your academic seminar series, and you can find it [here](#).

This database has been personalized for your department.

Sincerely,



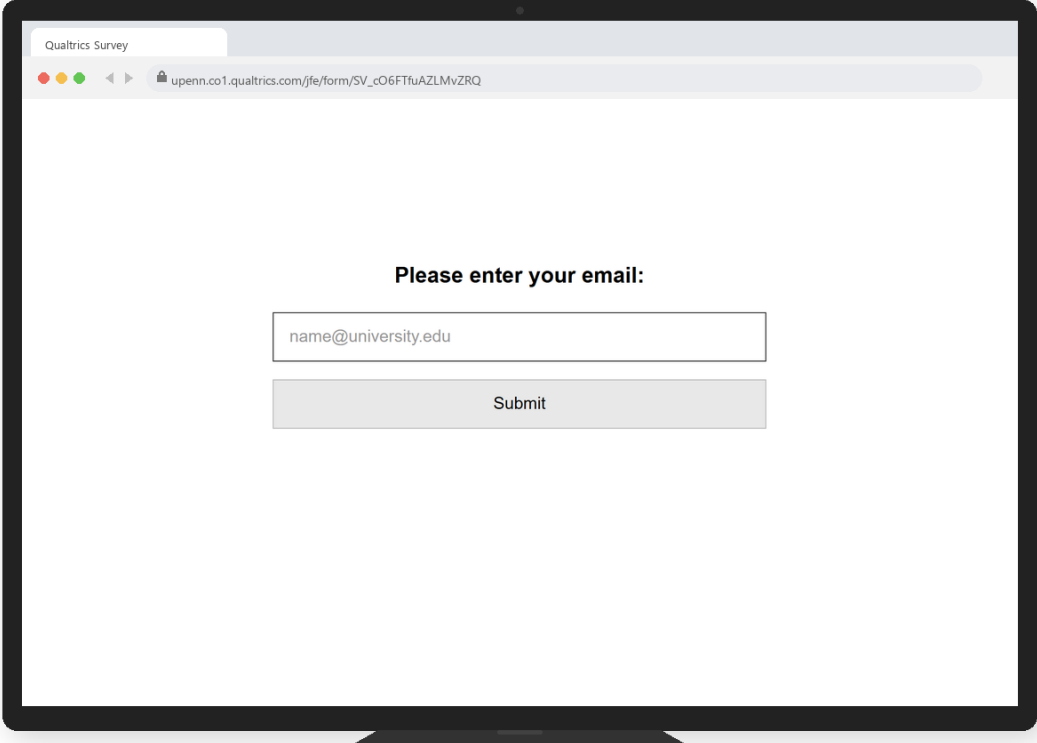
The Seminar Support Team
The University of Pennsylvania

P.S. We'd be grateful if you would **please refrain from sharing this email with colleagues at other universities, as it has been tailor-made for you** and sharing it would jeopardize our goal of assessing its impact (though you are welcome to share it with others who co-organize your department's seminars).

Notes: Sample email sent to study participants. The email varied along two dimensions. Recipients were addressed either as "Seminar Organizer" or as "Professor X" depending on whether they were identified as their department's seminar organizer or department chair, respectively (yellow highlighting). Recipients were either told they were being contacted in their capacity as the "current chair of your department" or "organizer of your [name of seminar series inserted here]" depending on whether they were identified as their department's chair or seminar organizer, respectively (yellow highlighting). The email also contained a key treatment manipulation: (1) the *URM Directory Treatment* condition shared "a database of scholars in your field who might enhance the diversity of your academic seminar series" (red text), while (2) the *Directory Control* condition shared "a database of departments in your field that you can search to enhance the diversity of your academic seminar series" (blue text). Treatment emails included a hyperlink to a curated directory of underrepresented minority faculty from peer institutions (see Figure S3); control emails included a hyperlink to a list of peer department websites (see Figure S4). All emails were sent from seminar-support@wharton.upenn.edu during the week of June 24, 2024, with follow-up reminders sent the week of July 9, 2024 to non-responders. This email has been anonymized by removing the recipient's surname from the salutation and their email address. No highlighted or colorful text appeared in the actual study emails. All highlights and color have been added here to emphasize what was varied in our experiment.

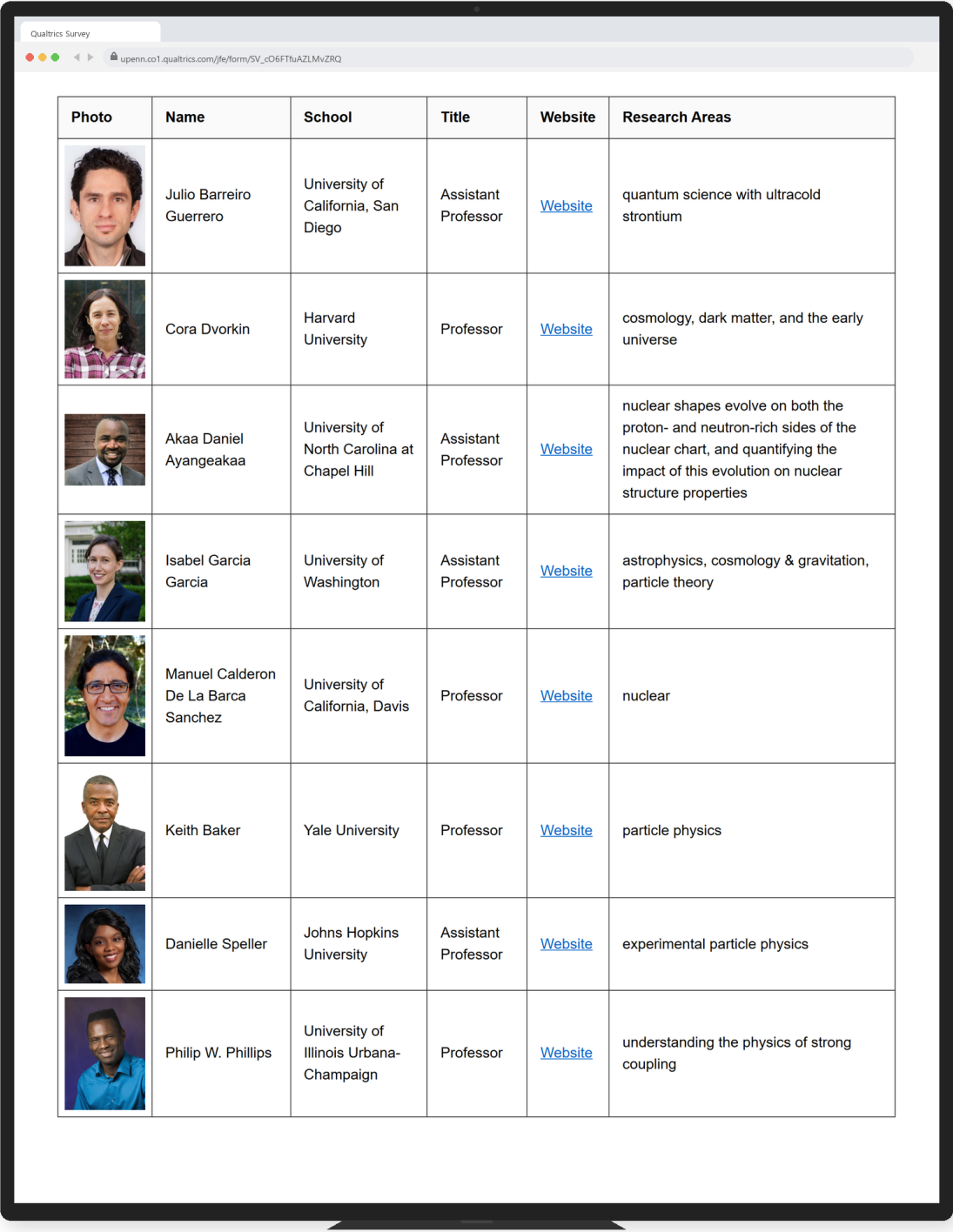
We sent at most 300 emails to participants per day in an attempt to avoid triggering institutional spam filters. Each person in our study received only one email regardless of how many seminars they oversaw. In cases where a department chair also served as a seminar organizer ($n=8$), we treated them as the department chair and sent them only the department chair email, so the 1,019 seminar organizers and 567 department chairs represent mutually exclusive groups. To reduce spillover across institutions, recipients were also asked not to forward our email to colleagues at other universities.

Figure S2. Directory Landing Page

A screenshot of a web browser displaying a Qualtrics survey landing page. The browser's address bar shows the URL "upenn.co1.qualtrics.com/jfe/form/SV_c06FTfuAZLMvZRQ". The page content is centered and features the text "Please enter your email:" above a text input field. The input field contains the placeholder text "name@university.edu". Below the input field is a grey "Submit" button. The entire page is framed by a black border, suggesting it is a monitor display.

Notes: The landing page displayed to both treatment and control condition study participants upon accessing their assigned directory link. Recipients were prompted to enter their university email address to verify their institutional affiliation before viewing their personalized directory. This verification step served to restrict directory access to intended recipients and reduce potential spillover across institutions.

Figure S3. URM Directory Treatment Interface

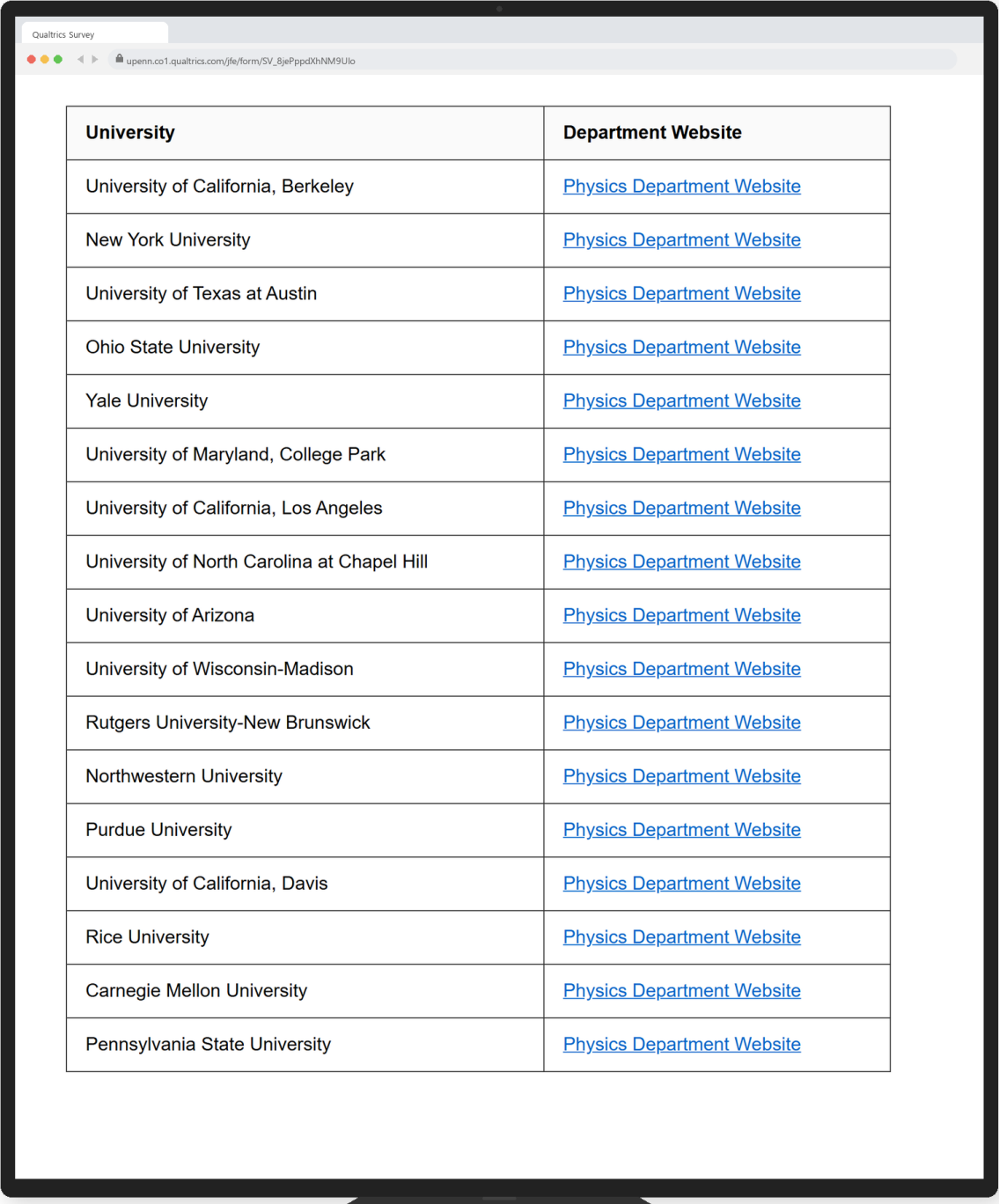


The image shows a tablet screen displaying a Qualtrics survey. The browser address bar shows the URL: upenn.co1.qualtrics.com/jfe/form/SV_c06FTfuAZLMvZRQ. The survey content is a table with six columns: Photo, Name, School, Title, Website, and Research Areas. The table lists eight URM faculty members from various institutions, including the University of California, Harvard University, University of North Carolina at Chapel Hill, University of Washington, University of California, Davis, Yale University, Johns Hopkins University, and University of Illinois Urbana-Champaign. Each entry includes a headshot, name, institution, title, a link to their website, and a brief description of their research areas.

Photo	Name	School	Title	Website	Research Areas
	Julio Barreiro Guerrero	University of California, San Diego	Assistant Professor	Website	quantum science with ultracold strontium
	Cora Dvorkin	Harvard University	Professor	Website	cosmology, dark matter, and the early universe
	Akaa Daniel Ayangeakaa	University of North Carolina at Chapel Hill	Assistant Professor	Website	nuclear shapes evolve on both the proton- and neutron-rich sides of the nuclear chart, and quantifying the impact of this evolution on nuclear structure properties
	Isabel Garcia Garcia	University of Washington	Assistant Professor	Website	astrophysics, cosmology & gravitation, particle theory
	Manuel Calderon De La Barca Sanchez	University of California, Davis	Professor	Website	nuclear
	Keith Baker	Yale University	Professor	Website	particle physics
	Danielle Speller	Johns Hopkins University	Assistant Professor	Website	experimental particle physics
	Philip W. Phillips	University of Illinois Urbana-Champaign	Professor	Website	understanding the physics of strong coupling

Notes: Sample view of a portion of the personalized directory that would have been provided to study participants in MIT's physics department in the *URM Directory Treatment* condition. The directory displayed at least 20 underrepresented minority (URM) faculty members from peer departments (ranked within 20 positions of the recipient's department in the 2024 U.S. News & World Report rankings). Each entry included the faculty member's photograph, name, institution, academic title, website link, and research areas.

Figure S4. Directory Control Interface

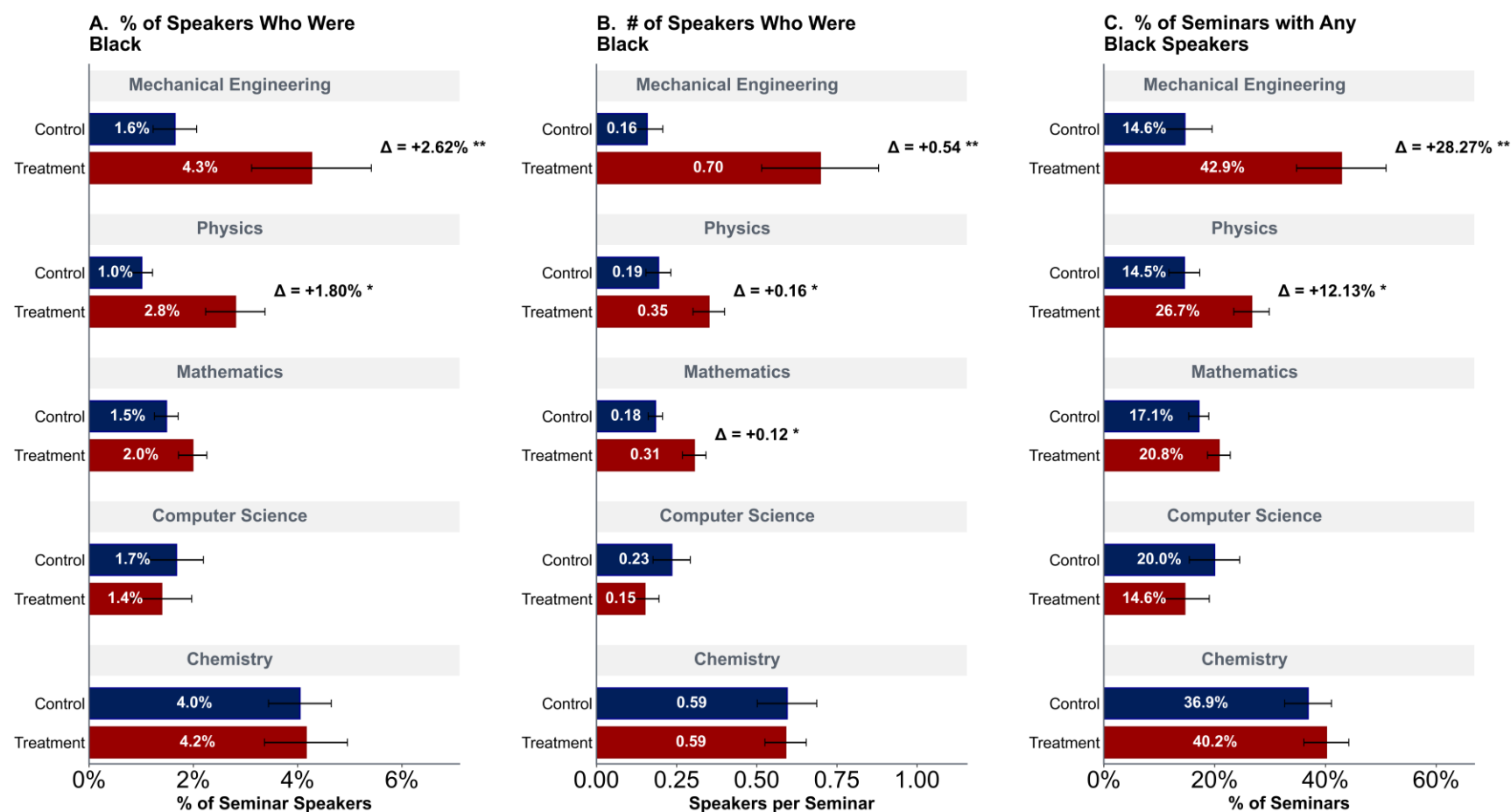


The image shows a screenshot of a web browser displaying a Qualtrics Survey. The browser's address bar shows the URL: upenn.co1.qualtrics.com/file/form/SV_8jePpdXhNM9Ulo. The survey content is a table with two columns: 'University' and 'Department Website'. The table lists 16 universities, each with a corresponding link to its physics department website. The links are all identical: 'Physics Department Website'.

University	Department Website
University of California, Berkeley	Physics Department Website
New York University	Physics Department Website
University of Texas at Austin	Physics Department Website
Ohio State University	Physics Department Website
Yale University	Physics Department Website
University of Maryland, College Park	Physics Department Website
University of California, Los Angeles	Physics Department Website
University of North Carolina at Chapel Hill	Physics Department Website
University of Arizona	Physics Department Website
University of Wisconsin-Madison	Physics Department Website
Rutgers University-New Brunswick	Physics Department Website
Northwestern University	Physics Department Website
Purdue University	Physics Department Website
University of California, Davis	Physics Department Website
Rice University	Physics Department Website
Carnegie Mellon University	Physics Department Website
Pennsylvania State University	Physics Department Website

Notes: Sample view of a portion of the directory that would have been provided to study participants in Stanford University's physics department in the *Directory Control* condition. The directory listed peer departments (identical to those from which URM faculty were drawn in the *URM Directory Treatment* condition) with direct links to each department's website. This design required participants to independently search for and identify potential speakers, maintaining the typical processes associated with finding candidates.

Figure S5. Treatment Effects on Black Speakers by Discipline



Notes: This figure displays the treatment effects of Black speaker outcomes. Panel A displays the regression-adjusted percentage of seminar speakers who were Black, Panel B displays regression-adjusted number of speakers per seminar who were Black, and Panel C displays the regression-adjusted percentage of seminars with at least one Black speaker, each disaggregated by discipline. Blue bars represent seminars in the *Directory Control* condition (n=862), while red bars represent seminars in the *URM Directory Treatment* condition (n=824). Values are regression-adjusted predictions from ordinary least squares models run separately by discipline. All models include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted), and (2)

randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted). Models also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Error bars represent +/- 1 standard error. Standard errors are clustered at the seminar series' department level.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Section 6: Supplementary Tables

Table S1. Analysis of Attrition in Academic Seminar Series Between the 2023-24 and 2024-25 Academic Years

Summary of How the Seminar Disappeared between 2023-24 and 2024-25	n	%	URM Directory Treatment	Directory Control
Maintained an active website but had no publicly listed speakers for 2024-25	43	22.1%	19 (17.6%)	24 (27.6%)
Removed or restricted public access to information about speakers for 2024-25 ^a	87	44.6%	55 (50.9%)	32 (36.8%)
Explicitly announced that it had been discontinued or was inactive for 2024-25 ^b	65	33.3%	34 (31.5%)	31 (35.6%)
TOTAL	195	100.0%	108 (100%)	87 (100.0%)

Notes: This table reports the breakdown of 195 academic seminar series (10.4% of the original sample) that could not be tracked during the 2024–25 academic year outcome measurement period. The original experimental sample included 1,881 seminar series identified during the 2023–24 academic year across 146 R1 universities. Of these, 1,686 (89.6%) maintained publicly accessible speaker information in 2024-25 and constitute the analysis sample. This table categorizes the 195 missing seminars based on a systematic review of their department websites and online presence conducted between August 2024 and June 2025. The difference in attrition rates between treatment and control conditions was not statistically significant at either the department level (7.4% vs. 5.2%; $B = 0.022$, $SE = 0.021$, $p = .284$, HC3-robust) or the seminar level (11.6% vs. 9.2%; $B = 0.024$, $SE = 0.016$, $p = .133$, clustered at the department level); see Table S5 for full results. Treatment and control columns show n (% within condition). Seminars were selected on the basis of their 2023-24 characteristics, and these same seminar series were then tracked during the 2024-25 academic year regardless of whether their format changed.

^a No public information includes seminars where the website was removed, locked behind university access, status uncertain, alternative links found without data, access restricted behind login, website errors, or otherwise no public information available.

^b Discontinued for 2024–25 includes seminars explicitly marked as discontinued, with the last recorded activity in 2023 or Spring 2024.

Table S2. Summary Statistics and Balance Checks (Analyzed Sample)

Characteristic	Mean Across Conditions	Mean in Directory Control	Mean in URM Directory Treatment	<i>p</i> -value (RI)
Panel A: Department-Level Characteristics (N = 532)				
# of Seminars Hosted by Department	3.17 (3.50)	3.18 (3.52)	3.16 (3.48)	0.745
Chemistry Department	22.93%	19.93%	26.05%	0.082†
Computer Science Department	15.41%	17.71%	13.03%	0.131
Mathematics Department	25.38%	26.57%	24.14%	0.480
Mechanical Engineering Department	12.41%	11.81%	13.03%	0.596
Physics Department	23.87%	23.99%	23.75%	0.994
Number of Faculty in Department	34.05 (18.04)	32.92 (15.49)	35.23 (20.31)	0.116
% Black Faculty in Department	1.63%	1.43%	1.84%	0.117
% Hispanic Faculty in Department	2.42%	2.26%	2.59%	0.341
% Female Faculty in Department	20.35%	19.99%	20.72%	0.282
% Department Chairs who are Black	2.26%	3.33%	1.15%	0.087†
% Department Chairs who are Hispanic	5.27%	5.93%	4.60%	0.465
% Department Chairs who are Female	17.89%	18.52%	17.24%	0.677
Joint F-test (RI p-value)	<i>F</i> (12, 512) = 1.66			0.108
Panel B: Seminar-Level Characteristics (N = 1,686)				
Number of Email Recipients	4.86 (4.46)	4.68 (4.25)	5.05 (4.67)	0.516
% Seminar Organizers who are Black	4.70%	5.65%	3.74%	0.519
% Seminar Organizers who are Hispanic	7.77%	7.14%	8.40%	0.634
% Seminar Organizers who are Female	43.59%	44.98%	42.18%	0.573
Joint F-test (RI p-value)	<i>F</i> (4, 1344) = 0.32			0.925

Notes: This table reports balance checks for our final analytic sample including 532 departments and 1,686 seminars. Panel A reports department-level characteristics with each department weighted equally. Panel B reports seminar-level characteristics. Each row reports condition-specific means for the pre-treatment characteristic listed in the row label. Following Athey and Imbens (2017), the p-value column reports randomization inference (RI) p-values from 5,000 within-strata permutations of the treatment assignment, based on the absolute t-statistic from an OLS regression of the characteristic on the treatment indicator with randomization-strata fixed effects. Panel A uses HC3-robust standard errors; Panel B clusters standard errors at the department level. The joint F-test at the bottom of each panel tests the null that all listed covariates are jointly unrelated to treatment assignment. Randomization inference follows Hansen and Bowers (2008) and preserves the original stratified randomization design. Note that the number of seminars used in stratification was based on the 1,881 seminars in our original sample and how many seminars a department hosted in the 2023–24 academic year, and was not based on the number of seminars a department hosted that continued into the 2024–25 academic year (this final analysis sample included 195 fewer seminars total). We stratified department randomization based on the number of seminars a department hosted, creating seven strata: 1 seminar (n = 280), 2-3 seminars (n = 280), 4-5 seminars (n = 300), 6-7 seminars (n = 278), 8-11 seminars (n = 298), 12-17 seminars (n = 318), and 18-28 seminars (n = 127). Within each stratum, departments were randomly assigned with equal probability to each condition.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Directory engagement. The average number of email link clicks per seminar was significantly higher in the URM Directory Treatment condition: participants clicked 0.74 times per seminar in the URM Directory Treatment condition compared to 0.63 times in the Directory Control condition, $t(1684) = 1.99$, $p = .047$.

Table S3. Characteristics of Speakers in STEM Seminars in the 2024–25 Academic Year, by Race/Ethnicity

	All	White	Asian	Hispanic	Black
<i>N</i> appearances	23,455	14,620	7,179	1,127	523
<i>N</i> unique speakers	17,712	10,999	5,391	887	430
Panel A: Speaker's Gender					
Woman	3,986 (17.0%)	2,578 (17.6%)	1,118 (15.6%)	198 (17.6%)	90 (17.2%)
Man	19,469 (83.0%)	12,042 (82.4%)	6,061 (84.4%)	929 (82.4%)	433 (82.8%)
Panel B: Speaker's Academic Rank					
Professor	6,605 (28.2%)	4,731 (32.4%)	1,507 (21.0%)	283 (25.1%)	84 (16.1%)
Associate Professor	2,109 (9.0%)	1,338 (9.2%)	633 (8.8%)	84 (7.5%)	53 (10.1%)
Assistant Professor	4,441 (18.9%)	2,413 (16.5%)	1,708 (23.8%)	209 (18.5%)	109 (20.8%)
Post-doc	2,622 (11.2%)	1,473 (10.1%)	956 (13.3%)	135 (12.0%)	56 (10.7%)
Graduate Student	3,080 (13.1%)	1,594 (10.9%)	1,204 (16.8%)	182 (16.1%)	99 (18.9%)
Non-faculty	4,598 (19.6%)	3,071 (21.0%)	1,171 (16.3%)	234 (20.8%)	122 (23.3%)
Panel C: Speaker's Institutional Affiliation					
Higher-ranked (external)	6,891 (29.4%)	4,086 (27.9%)	2,359 (32.9%)	290 (25.7%)	156 (29.8%)
Lower-ranked (external)	4,045 (17.2%)	2,492 (17.0%)	1,297 (18.1%)	183 (16.2%)	73 (14.0%)
Host institution	5,085 (21.7%)	3,100 (21.2%)	1,583 (22.1%)	250 (22.2%)	150 (28.7%)
Not in U.S. News rankings	7,434 (31.7%)	4,942 (33.8%)	1,940 (27.0%)	404 (35.8%)	144 (27.5%)
At Institution in Peer Set	4,840 (20.6%)	2,874 (19.7%)	1,673 (23.3%)	192 (17.0%)	101 (19.3%)
Not at Institution in Peer Set	18,615 (79.4%)	11,746 (80.3%)	5,506 (76.7%)	935 (83.0%)	422 (80.7%)

Notes: This table describes the characteristics of the speakers who appeared across 1,686 STEM seminars in our analytic sample during the 2024–25 academic year, broken down by speaker race/ethnicity. The unit of observation is a speaker appearance (a speaker may appear in multiple seminars and would then be counted repeatedly here). Entries show the count and column percentage within each racial/ethnic group. Academic rank is as reported in public institutional profiles. Institutional affiliation categorizes speakers relative to the host department's 2024 U.S. News & World Report discipline-specific ranking. Higher-ranked (external) includes speakers from institutions ranked above the host; lower-ranked (external) includes those ranked below. Host institution includes speakers whose affiliation matches the seminar's host university. Unranked includes speakers from international, industry, or other affiliations not in our ranking data. At Institution in Peer Set includes speakers whose institutional affiliation is one of the approximately 40 peer departments ranked within +/- 20 positions of the host department—the same set of institutions eligible for inclusion in the URM directory (see Table 3 in the main text). Not at Institution in Peer Set includes all remaining speakers (those at the host institution, at unranked institutions, or at ranked institutions outside the peer set). Race/ethnicity and gender were coded using a multi-method approach combining facial recognition, name-based classification, and manual verification (see Methods section for details).

Table S4. Summary Statistics and Balance Checks (Full Randomized Sample)

Characteristic	Mean Across Conditions	Mean in Directory Control	Mean in URM Directory Treatment	p-value (RI)
Panel A: Department-Level Characteristics (N = 568)				
# of Seminars Hosted by Department	3.31 (3.74)	3.32 (3.74)	3.30 (3.75)	0.885
Chemistry Department	22.54%	19.58%	25.53%	0.097†
Computer Science Department	16.55%	18.18%	14.89%	0.298
Mathematics Department	24.12%	25.52%	22.70%	0.423
Mechanical Engineering Department	13.91%	13.29%	14.54%	0.639
Physics Department	22.89%	23.43%	22.34%	0.747
Number of Faculty in Department	33.57 (17.82)	32.42 (15.36)	34.73 (19.97)	0.102
% Black Faculty in Department	1.61%	1.41%	1.82%	0.099†
% Hispanic Faculty in Department	2.50%	2.30%	2.71%	0.262
% Female Faculty in Department	20.19%	19.82%	20.56%	0.256
% Department Chairs who are Black	2.12%	3.16%	1.06%	0.076†
% Department Chairs who are Hispanic	5.29%	5.61%	4.96%	0.773
% Department Chairs who are Female	17.28%	17.89%	16.67%	0.664
Joint F-test (RI p-value)	F(12, 549) = 1.56			0.154
Panel B: Seminar-Level Characteristics (N = 1,881)				
Number of Email Recipients	4.82 (4.41)	4.65 (4.22)	4.99 (4.59)	0.498
% Seminar Organizers who are Black	4.80%	5.89%	3.73%	0.555
% Seminar Organizers who are Hispanic	7.74%	7.49%	7.99%	0.885
% Seminar Organizers who are Female	43.35%	44.32%	42.38%	0.638
Joint F-test (RI p-value)	F(4, 1505) = 0.30			0.932

Notes: This table reports balance checks for the full randomized sample (568 departments, 1,881 seminars), including the 36 departments (195 seminars) that attrited from the final analytic sample. Panel A reports department-level characteristics with each department weighted equally. Panel B reports seminar-level characteristics. Each row reports condition-specific means for the pre-treatment characteristic listed in the row label. Following Athey and Imbens (2017), the p-value column reports randomization inference (RI) p-values from 5,000 within-strata permutations of the treatment assignment, based on the absolute t-statistic from an OLS regression of the characteristic on the treatment indicator with randomization-strata fixed effects. Panel A uses HC3-robust standard errors; Panel B clusters standard errors at the department level. The joint F-test at the bottom of each panel tests the null that all listed covariates are jointly unrelated to treatment assignment. Email recipients were distributed across disciplines as follows: 42.2% worked in Mathematics departments, 20.1% worked in Physics departments, 16.1% worked in Chemistry departments, 13.0% worked in Computer Science departments, and 8.7% worked in Mechanical Engineering departments. Randomization inference follows Hansen and Bowers (2008) and preserves the original stratified randomization design.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S5. Tests for Differential Attrition of Seminars and Departments across Conditions

	Did All Seminars in the Department Disappear between Treatment Assignment and Outcome Measurement? (1)	Did the Seminar Disappear between Treatment Assignment and Outcome Measurement? (2)
URM Directory Treatment	0.0220 (0.0205)	0.0242 (0.0161)
p-value	0.284	0.133
Control attrition rate	5.2%	9.2%
Treatment attrition rate	7.4%	11.6%
N	568	1,881

Notes: This table tests for differential attrition by treatment condition. Column (1) reports a department-level OLS regression where the outcome variable is a binary indicator for whether the department in question dropped out of the analytic sample. The primary predictor is a *URM Directory Treatment* indicator. The regression includes HC3-robust standard errors (N = 568 departments). Column (2) reports a seminar-level OLS regression where the outcome variable is a binary indicator for whether the seminar series in question dropped out of the analytic sample. The primary predictor is a *URM Directory Treatment* indicator. The regression includes standard errors clustered at the department level (N = 1,881 seminars).

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S6. Intent-to-Treat (ITT) Analysis with Imputation of Outcomes in Unobserved Seminars

	DV: % of AY24-25 Seminars Speakers Who Were Black		DV: Number of AY24-25 Seminars Speakers Who Were Black		DV: Did AY24-25 Seminar Have at Least One Black Speaker?	
	(1)	(2)	(3)	(4)	(5)	(6)
URM Directory Treatment	0.5908*	0.5920*	0.0657†	0.0656*	0.0439*	0.0438*
	(0.2742)	(0.2633)	(0.0347)	(0.0332)	(0.0201)	(0.0202)
# of Faculty in Dept. of Seminar		-0.0147*		-0.0019*		-0.0011†
		(0.0060)		(0.0009)		(0.0006)
% of Faculty in Dept. of Seminars Who Are Female		2.2260		0.3049		0.1956
		(1.8111)		(0.2540)		(0.1512)
% of Faculty in Dept. of Seminars Who Are URMs		6.1283		0.4193		0.2409
		(4.2155)		(0.5193)		(0.2810)
U.S. News Ranking of Dept. of Seminar		0.0094		0.0012		-0.0000
		(0.0061)		(0.0008)		(0.0004)
# of Faculty Displayed in URM Directory for Dept. of Seminar		0.0553*		0.0086**		0.0051**
		(0.0264)		(0.0033)		(0.0019)
# of Peer Departments with 20+ URM Faculty		0.1725		0.0083		0.0051
		(0.1402)		(0.0114)		(0.0061)
# of Dept Reps Who Received Treatment Email		0.0775**		0.0128**		0.0093***
		(0.0290)		(0.0040)		(0.0025)
Observations	1,881	1,881	1,881	1,881	1,881	1,881
Adjusted R ²	0.031	0.037	0.051	0.062	0.044	0.054

Notes: This table re-estimates treatment effects on the full randomized sample of 1,881 seminars, imputing zeros for Black-speaker outcomes in the 195 seminars that were not observable in 2024–25. This conservative specification assumes that unobserved seminars featured no Black speakers. Models 1–2 predict the percentage of Black speakers in a given seminar series in the 2024–25 academic year. Models 3–4 predict the number of Black speakers in a given seminar series. Models 5–6 predict whether a given seminar series featured at least one Black speaker. Models 1, 3, and 5 include fixed effects for (1) the seminar’s discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series’ organizers (here we include nine fixed effects for different email batches sent between June 24–28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023–24 academic year (1 seminar series, 2–3 seminar series, 4–5 seminar series, 6–7 seminar series, 8–11 seminar series, 12–17 seminar series; an indicator for hosting 18–28 seminar series is omitted). Models 2, 4, and 6 include the same fixed effects but also include controls for (1) the total number of faculty in the seminar’s host department in summer 2024, (2) the fraction of faculty in the seminar’s host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar’s host department who were women in summer 2024, (4) the seminar’s host department’s U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university’s overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series’ department level, are shown in parentheses.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S7. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation in STEM Seminars in the 2024-25 Academic Year by Semester

	DV: % of Seminar Speakers Who Were Black				DV: Number of Seminar Speakers Who Were Black				DV: Did Seminar Have at Least One Black Speaker?			
	<i>Fall '24 Semester</i>		<i>Spring '25 Semester</i>		<i>Fall '24 Semester</i>		<i>Spring '25 Semester</i>		<i>Fall '24 Semester</i>		<i>Spring '25 Semester</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
URM Directory Treatment	0.3946 (0.2960)	0.4147 (0.2873)	0.8694* (0.3878)	0.8913* (0.3755)	0.0458* (0.0186)	0.0459* (0.0183)	0.0357 (0.0291)	0.0348 (0.0280)	0.0373* (0.0148)	0.0380* (0.0151)	0.0271 (0.0190)	0.0271 (0.0188)
# of Faculty in Dept. of Seminar		-0.0142* (0.0072)		-0.0163† (0.0088)		-0.0006 (0.0005)		-0.0014* (0.0007)		-0.0006 (0.0005)		-0.0010† (0.0005)
% of Faculty in Dept. of Seminar Who Are Female		2.0216 (1.7778)		2.4117 (2.6715)		0.0268 (0.1303)		0.2269 (0.2138)		0.0026 (0.1081)		0.1910 (0.1486)
% of Faculty in Dept. of Seminar Who Are URM		5.8266 (4.2557)		3.4763 (5.6748)		0.3594 (0.2998)		0.1469 (0.3949)		0.3391 (0.2521)		0.0824 (0.2594)
U.S. News Ranking of Dept. of Seminar		0.0226** (0.0074)		0.0055 (0.0084)		0.0012** (0.0004)		0.0003 (0.0007)		0.0007* (0.0003)		-0.0000 (0.0004)
# of Faculty Displayed in URM Directory for Dept. of Seminar		0.1232*** (0.0312)		0.0234 (0.0351)		0.0075*** (0.0018)		0.0016 (0.0027)		0.0059*** (0.0014)		0.0015 (0.0018)
# of Peer Departments with 20+ URM Faculty		0.2354 (0.1881)		0.1353 (0.1830)		0.0034 (0.0062)		0.0088 (0.0103)		0.0026 (0.0054)		0.0053 (0.0065)
# of Dept Reps Who Received Treatment Email		0.0837* (0.0354)		0.0727† (0.0441)		0.0047* (0.0024)		0.0090** (0.0032)		0.0040* (0.0019)		0.0069** (0.0024)
Observations	1,686	1,686	1,686	1,686	1,686	1,686	1,686	1,686	1,686	1,686	1,686	1,686
Adjusted R ²	0.015	0.033	0.013	0.012	0.023	0.036	0.028	0.037	0.020	0.032	0.022	0.025

Notes: This table reports the results of twelve OLS regressions estimating the effect of random assignment to the URM Directory Treatment condition on Black speaker representation, disaggregated by semester. Models 1–2 predict the percentage of Black speakers in the fall 2024 semester; Models 3–4 predict the percentage in the spring 2025 semester; Models 5–6 predict the number of Black speakers in the fall 2024 semester; Models 7–8 predict the number of Black speakers in the spring 2025 semester; Models 9–10 predict whether a seminar featured at least one Black speaker in the fall 2024 semester; Models 11–12 predict whether a seminar featured at least one Black speaker in the spring 2025 semester. Within each pair of regressions, odd-numbered models include only fixed effects for (1) the seminar’s discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series’ organizers (here we include nine fixed effects for different email batches sent between June 24–28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023–24 academic year (1 seminar series, 2–3 seminar series, 4–5 seminar series, 6–7 seminar series, 8–11 seminar series, 12–17 seminar series; an indicator for hosting 18–28 seminar series is omitted). Even-numbered models include the same fixed effects but also include controls for (1) the total number of

faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. For seminars with no speakers in a given semester (Fall: 508; Spring: 308), percentage outcomes are coded as zero. Results are substantively unchanged when these seminars are excluded. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses. For consistency across institutions, we define "Fall semester" as seminars occurring between August 1 and December 31, 2024, and "Spring semester" as seminars occurring between January 1 and June 30, 2025, regardless of whether the institution uses a semester, trimester, or quarter system. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

As expected given the null effects overall, semester-by-semester analyses of overall URM speaker and Hispanic speaker representation revealed no significant effects (all p 's $> .10$; see Extended Tables E3–E4 on OSF).

6.1 Confidence Sets for Ranks of Discipline-Specific Treatment Effects

We applied the confidence sets for ranks procedure developed by Mogstad et al. (2024) to formally characterize the ranking of discipline-specific treatment effects. Marginal, one-sided, 95% confidence intervals were constructed for the rank of each discipline's treatment effect based on the Treatment x Discipline interaction coefficients reported in Table S8 below. When estimating the impact of our treatment on the percentage of Black speakers in a given seminar over the year post-intervention, Mechanical Engineering and Physics are the only disciplines whose 95% confidence set for rank includes 1 (i.e., the top position). When estimating the impact of our treatment on the number of Black speakers in a given seminar over the year post-intervention and the probability of a seminar featuring any Black speaker in the year post-intervention, Mechanical Engineering is definitively the top-ranked discipline. Table S9 reports these confidence sets.

Table S8. Linear Regressions Estimating Heterogeneity in Treatment Effects on Black Speaker Representation by Academic Discipline in the 2024-25 Academic Year

	DV: % of AY24-25 Seminar Speakers Who Were Black		DV: Number of AY24-25 Seminar Speakers Who Were Black		DV: Did AY24-25 Seminar Have at Least One Black Speaker?	
	(1)	(2)	(3)	(4)	(5)	(6)
URM Directory Treatment x Mechanical Engineering	3.8488*** (1.1653)	3.8227** (1.1909)	0.5813** (0.1878)	0.5740** (0.1814)	0.3210** (0.0984)	0.3375*** (0.1010)
URM Directory Treatment x Physics	1.6506** (0.6309)	1.6345* (0.6495)	0.1451* (0.0655)	0.1433* (0.0675)	0.0924* (0.0446)	0.0932* (0.0461)
URM Directory Treatment x Chemistry	0.4932 (1.0210)	0.4534 (0.9276)	-0.0500 (0.1272)	-0.0522 (0.1159)	0.0607 (0.0667)	0.0634 (0.0627)
URM Directory Treatment x Mathematics	0.2254 (0.3872)	0.1948 (0.3806)	0.0767 (0.0499)	0.0686 (0.0490)	0.0270 (0.0292)	0.0203 (0.0283)
URM Directory Treatment x Computer Science	-0.2053 (0.8615)	0.1502 (0.8474)	-0.0727 (0.0890)	-0.0272 (0.0881)	-0.0379 (0.0762)	-0.0253 (0.0732)
Mechanical Engineering	-0.2903 (0.7289)	-12.1672 [†] (7.1803)	-0.0595 (0.0931)	-1.0360 [†] (0.5945)	0.0149 (0.0740)	-0.6639* (0.3126)
Physics	-0.0403 (0.4938)	-12.1809 [†] (7.2239)	0.0806 (0.0681)	-0.8982 (0.5923)	0.0870* (0.0430)	-0.5577 [†] (0.3122)
Chemistry	2.6141** (0.8724)	-8.9020 (6.9897)	0.4815*** (0.1201)	-0.4253 (0.5717)	0.2739*** (0.0621)	-0.3316 (0.3041)
Mathematics	0.8435* (0.4236)	-11.2574 (7.0025)	0.1399* (0.0599)	-0.8339 (0.5768)	0.1346** (0.0429)	-0.4850 (0.3037)
Computer Science	0.4117 (0.7086)	-11.1433 (7.1328)	0.0739 (0.0925)	-0.8383 (0.5747)	0.0989 (0.0654)	-0.5221 [†] (0.3064)
# of Faculty in Dept. of Seminar		-0.0146* (0.0067)		-0.0020* (0.0010)		-0.0010 (0.0006)
% of Faculty in Dept. of Seminar Who Are Female		2.3354 (1.9308)		0.3012 (0.2728)		0.1921 (0.1638)
% of Faculty in Dept. of		5.7742		0.4434		0.1875

Seminar Who Are URM	(4.6080)	(0.5777)	(0.3057)
U.S. News Ranking of	0.0112 [†]	0.0014	0.0001
Dept. of Seminar	(0.0065)	(0.0009)	(0.0004)
# of Faculty Displayed in	0.0628*	0.0094**	0.0059**
URM Directory for Dept.	(0.0286)	(0.0035)	(0.0020)
of Seminar			
# of Peer Departments with	0.2275	0.0133	0.0084
20+ URM Faculty	(0.1656)	(0.0133)	(0.0070)
# of Dept Reps Who	0.0870**	0.0136**	0.0101***
Received Treatment Email	(0.0311)	(0.0044)	(0.0027)
Observations	1,686	1,686	1,686
Adjusted R ²	0.144	0.151	0.216
		0.227	0.259
			0.268

Notes: This table reports the results of six OLS regressions estimating treatment-by-discipline interaction effects on Black speaker representation, where the primary predictors in each model are the interactions between treatment assignment and academic discipline indicators. All models suppress the intercept and include indicators for all five disciplines studied (mathematics, physics, chemistry, computer science, and mechanical engineering) and treatment-by-discipline interaction terms for each discipline studied. Interaction terms are ordered from largest to smallest treatment effect estimate. Models 1, 3, and 5 include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted) and (2) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted). Models 2, 4, and 6 include the same fixed effects but also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URM in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses.

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S9. Confidence Sets for Rankings of Discipline-Specific Treatment Effects on Black Speaker Representation in the 2024-25 Academic Year

	DV: % of Seminar Speakers Who Were Black		DV: Number of Seminar Speakers Who Were Black		DV: Did Seminar Feature at Least One Black Speaker?	
	(1)	Lower Bound	(2)	Lower Bound	(3)	Lower Bound
URM Directory Treatment × Mechanical Engineering	3.823**	1	0.574**	1	0.338***	1
	(1.191)		(0.181)		(0.101)	
URM Directory Treatment × Physics	1.635*	1	0.143*	2	0.093*	2
	(0.649)		(0.067)		(0.046)	
URM Directory Treatment × Chemistry	0.453	2	-0.052	2	0.063	2
	(0.928)		(0.116)		(0.063)	
URM Directory Treatment × Mathematics	0.195	3	0.069	2	0.020	2
	(0.381)		(0.049)		(0.028)	
URM Directory Treatment × Computer Science	0.150	2	-0.027	2	-0.025	2
	(0.847)		(0.088)		(0.073)	
Observations	1,686		1,686		1,686	
Adjusted R ²	0.151		0.227		0.268	

Notes: This table reports the results of three OLS regressions estimating discipline-specific treatment effects on Black speaker representation, along with the lower bounds of one-sided, 95% confidence sets for each discipline's rank, following the ranking inference methodology of Mogstad, Romano, Shaikh, and Wilhelm (2024). A lower bound of 1 (shown in bold) indicates that the discipline cannot be excluded from being the true top-ranked discipline at the 95% confidence level. Model 1 predicts the percentage of Black speakers in a given seminar series in the 2024-25 academic year. Model 2 predicts the number of Black speakers in a given seminar series in the 2024-25 academic year. Model 3 predicts whether a given seminar series featured at least one Black speaker in the 2024-25 academic year. All models suppress the intercept and include indicators for all five disciplines (chemistry, computer science, mathematics, mechanical engineering, and physics) and treatment-by-discipline interaction terms for each discipline. All models include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24–28, 2024; the tenth batch is omitted) and (2) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2–3 seminar series, 4–5 seminar series, 6–7 seminar series, 8–11 seminar series, 12–17 seminar series; an indicator for hosting 18–28 seminar series is omitted), and also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S10. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation in STEM Seminars by Academic Discipline (Separate Regressions) in the 2024-25 Academic Year

	DV: % of AY24-25 Seminar Speakers Who Were Black					DV: Number of AY24-25 Seminar Speakers Who Were Black					DV: Did AY24-25 Seminar Have at Least One Black Speaker?				
	<i>Math</i>	<i>Phys.</i>	<i>CS</i>	<i>Chem.</i>	<i>M.E.</i>	<i>Math</i>	<i>Phys.</i>	<i>CS</i>	<i>Chem.</i>	<i>M.E.</i>	<i>Math</i>	<i>Phys.</i>	<i>CS</i>	<i>Chem.</i>	<i>M.E.</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
URM Directory Treatment	0.5052 (0.3865)	1.7982** (0.6662)	-0.2852 (0.8233)	0.1176 (0.8979)	2.6193* (1.1751)	0.1211* (0.0540)	0.1571* (0.0629)	-0.0840 (0.0823)	-0.0044 (0.1059)	0.5403* (0.2230)	0.0364 (0.0263)	0.1213* (0.0472)	-0.0540 (0.0746)	0.0330 (0.0646)	0.2827* (0.1092)
# of Faculty in Dept. of Seminar	-0.0355*** (0.0097)	-0.0317 (0.0222)	0.0082 (0.0203)	0.0092 (0.0386)	-0.0120 (0.0314)	-0.0047*** (0.0013)	-0.0016 (0.0028)	0.0011 (0.0018)	0.0084 (0.0064)	-0.0059 (0.0048)	-0.0028** (0.0009)	-0.0019 (0.0023)	0.0009 (0.0017)	0.0074* (0.0033)	-0.0021 (0.0029)
% of Faculty in Dept. of Seminar Who Are Female	0.6275 (2.5146)	6.3950 (5.2173)	-3.0542 (4.4537)	10.8210† (6.1762)	11.8534 (10.2317)	0.1410 (0.3066)	0.0475 (0.5856)	-0.4183 (0.6201)	1.1469 (0.8799)	0.5617 (1.4417)	0.1901 (0.2043)	0.3481 (0.4425)	-0.3493 (0.4786)	0.7853 (0.4895)	1.2254 (0.8874)
% of Faculty in Dept. of Seminar Who Are URM	12.7952 (7.9187)	-22.5320** (8.5205)	16.4033 (11.8043)	26.2511† (14.1340)	10.0459 (13.2532)	1.3449 (1.1704)	-1.4807* (0.6804)	0.0378 (1.4861)	2.7893† (1.5624)	-2.1961 (1.9582)	0.1733 (0.4975)	-1.3643** (0.4923)	0.7958 (1.2160)	2.2395** (0.7839)	0.0432 (1.1186)
U.S. News Ranking of Dept. of Seminar	0.0069 (0.0084)	-0.0249 (0.0204)	-0.0081 (0.0470)	0.0317* (0.0126)	0.0636 (0.0433)	-0.0006 (0.0009)	-0.0067* (0.0031)	-0.0035 (0.0030)	0.0039† (0.0023)	0.0027 (0.0076)	-0.0005 (0.0006)	-0.0042* (0.0020)	-0.0031 (0.0028)	0.0019 (0.0013)	0.0049 (0.0031)
# of Faculty Displayed in URM Directory for Dept. of Seminar	0.0982† (0.0574)	-0.0100 (0.0730)	0.1143 (0.2340)	0.1423 (0.0976)	0.0623 (0.0738)	0.0049 (0.0079)	-0.0174† (0.0101)	-0.0086 (0.0160)	0.0280* (0.0129)	0.0076 (0.0126)	0.0080* (0.0040)	-0.0074 (0.0063)	-0.0054 (0.0152)	0.0136* (0.0066)	0.0098 (0.0059)
# of Peer Departments with 20+ URM Faculty	0.6701 (0.4938)	0.1722 (0.2020)	0.1490 (0.1746)		1.2536† (0.6855)	0.0261 (0.0408)	0.0037 (0.0125)	0.0480† (0.0289)		0.1474† (0.0881)	0.0114 (0.0133)	0.0082 (0.0103)	0.0277† (0.0166)		0.1743** (0.0605)
# of Dept Reps Who Received Treatment Email	0.0993** (0.0308)	-0.0691 (0.2042)	0.1293 (0.4460)	1.1100† (0.6095)	-2.0841 (1.3313)	0.0132*** (0.0036)	0.0056 (0.0253)	-0.0143 (0.0336)	0.1147† (0.0669)	-0.3344† (0.1807)	0.0105*** (0.0026)	0.0124 (0.0196)	-0.0086 (0.0310)	0.0427 (0.0378)	-0.2007* (0.0924)
Observations	823	361	143	277	82	823	361	143	277	82	823	361	143	277	82
Adjusted R ²	0.039	0.015	-0.081	-0.012	0.142	0.026	0.052	0.012	0.084	0.248	0.022	0.035	0.005	0.117	0.169

Notes: This table reports the results of fifteen OLS regressions estimating the effect of treatment on Black speaker representation, estimated separately for each of the five academic disciplines (mathematics, physics, computer science, chemistry, and mechanical engineering). Models 1-5 predict the percentage of Black speakers in a seminar series in

the 2024-25 Academic Year; Models 6-10 predict the number of Black speakers in a seminar series in the 2024-25 Academic Year; Models 11-15 predict whether a department's seminar series featured at least one Black speaker in the 2024-25 Academic Year. All models include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted) and (2) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted), and also include controls for (3) the total number of faculty in the seminar's host department in summer 2024, (4) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (5) the fraction of faculty in the seminar's host department who were women in summer 2024, (6) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (7) an indicator for missing department ranking, (8) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (9) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (10) the number of recipients of our experimental email about the target seminar series. Models 4, 9, and 14 (Chemistry) exclude the control for the total number of peer departments because all Chemistry seminars share the same value for this variable (40 peer departments), leaving no within-discipline variation. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

In the Directory Control condition, Black speakers comprised only 0.94% of speakers in Mechanical Engineering seminars and 1.07% in Physics seminars, compared to 3.88% in Chemistry seminars and 1.60% in Computer Science seminars. Given that we did not find any main effect of treatment on URM or Hispanic representation overall, it is not surprising that we also find no heterogeneity across disciplines for these groups (all p 's $> .15$; see Extended Tables E5–E6 on OSF).

Table S11. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation in STEM Seminars as a Function of Black Faculty Representation in the Seminar's Host Department in the 2024-25 Academic Year

	DV: % of AY24-25 Seminar Speakers Who Were Black		DV: Number of AY24-25 Seminar Speakers Who Were Black		DV: Did AY24-25 Seminar Have at Least One Black Speaker?	
	(1)	(2)	(3)	(4)	(5)	(6)
URM Directory Treatment	0.6902* (0.2970)	0.7307* (0.2879)	0.0785* (0.0374)	0.0786* (0.0356)	0.0533* (0.0218)	0.0545* (0.0212)
Mean-Centered # of Black Faculty in Dept. of Seminar	0.1694 (0.2268)	0.1927 (0.2103)	0.0196 (0.0311)	0.0223 (0.0277)	0.0262 (0.0206)	0.0262 (0.0190)
URM Directory Treatment × Mean-Centered # of Black Faculty in Dept. of Seminar	0.9577* (0.4728)	1.0235* (0.4712)	0.1117† (0.0613)	0.1254* (0.0596)	0.0484† (0.0289)	0.0566* (0.0280)
# of Faculty in Dept. of Seminar		-0.0177** (0.0067)		-0.0023* (0.0010)		-0.0013* (0.0006)
% of Faculty in Dept. of Seminar Who Are Female		1.9192 (1.8956)		0.2540 (0.2616)		0.1598 (0.1627)
% of Hispanic Faculty in Dept. of Seminar		-3.4496 (3.8196)		-1.1864* (0.5034)		-0.5794† (0.3050)
% of Native American Faculty in Dept. of Seminar		0.2491 (11.6194)		-0.3342 (1.1270)		-0.0740 (0.6658)
U.S. News Ranking of Dept. of Seminar		0.0121† (0.0065)		0.0016† (0.0009)		0.0001 (0.0004)
# of Faculty Displayed in URM Directory for Dept. of Seminar		0.0571† (0.0302)		0.0091* (0.0037)		0.0055** (0.0021)
# of Peer Departments with 20+ URM Faculty		0.2065 (0.1650)		0.0103 (0.0131)		0.0062 (0.0070)
# of Dept Reps Who Received Treatment Email		0.1151*** (0.0342)		0.0182*** (0.0049)		0.0122*** (0.0028)
Observations	1686	1686	1686	1686	1686	1686
Adjusted R ²	0.033	0.040	0.055	0.072	0.044	0.056

Notes: This table reports the results of six OLS regressions estimating the interactive effect of treatment assignment and the number of Black faculty in the seminar's host department on Black speaker representation. The number of Black faculty in a given department is mean-centered so that the treatment main effect is interpretable as the treatment effect at the average level of Black faculty representation. Models 1, 3, and 5 include fixed effects for (1) the seminar's discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted). Models 2, 4, and 6 include the same fixed effects but also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were Hispanic in summer 2024, (3) the fraction who were Native American in summer 2024, (4) the fraction of faculty in the seminar's host department who were women in summer 2024, (5) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (6) an indicator for missing department ranking, (7) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (8) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (9) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Consistent with the lack of robust main effects reported in the main text, we found no significant heterogeneity in the effect of treatment on overall URM or Hispanic speaker representation as a function of same-group faculty representation in the seminar's host department (see Extended Tables E11–E12 on OSF).

Table S12. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation by Whether the Seminar Organizer Is Black

	Seminars Without a Black Organizer			Seminars With a Black Organizer		
	DV: % of AY24-25 Seminar Speakers Who Were Black	DV: Number of AY24-25 Seminar Speakers Who Were Black	DV: Did AY24-25 Seminar Have at Least One Black Speaker?	DV: % of AY24-25 Seminar Speakers Who Were Black	DV: Number of AY24-25 Seminar Speakers Who Were Black	DV: Did AY24-25 Seminar Have at Least One Black Speaker?
	(1)	(2)	(3)	(4)	(5)	(6)
URM Directory Treatment	0.7417*	0.0756 [†]	0.0505*	-0.5676	-0.0812	-0.0035
	(0.3166)	(0.0394)	(0.0236)	(1.3451)	(0.1843)	(0.1123)
# of Faculty in Dept. of Seminar	-0.0156*	-0.0021 [†]	-0.0011	-0.0570 [†]	-0.0041	-0.0063*
	(0.0078)	(0.0011)	(0.0007)	(0.0340)	(0.0044)	(0.0030)
% of Faculty in Dept. of Seminar Who Are Female	1.5760	0.1782	0.1214	18.7832	2.6439 [†]	1.2640
	(2.1416)	(0.2977)	(0.1784)	(11.4886)	(1.5668)	(0.9464)
% of Faculty in Dept. of Seminar Who Are URM	6.2632	0.4957	0.2659	33.2323	0.6576	3.1692*
	(5.0019)	(0.5974)	(0.3199)	(21.2787)	(2.8610)	(1.4716)
U.S. News Ranking of Dept. of Seminar	0.0120 [†]	0.0015	0.0000	0.0176	0.0047	0.0016
	(0.0070)	(0.0009)	(0.0005)	(0.0211)	(0.0032)	(0.0021)
# of Faculty Displayed in URM Directory for Dept. of Seminar	0.0585 [†]	0.0086*	0.0054*	-0.0324	0.0063	0.0003
	(0.0319)	(0.0039)	(0.0022)	(0.1049)	(0.0150)	(0.0097)
# of Peer Departments with 20+ URM Faculty	0.2352	0.0142	0.0090	0.2786	0.0924	0.0100
	(0.1742)	(0.0140)	(0.0074)	(0.4599)	(0.0737)	(0.0299)
# of Dept Reps Who Received Treatment Email	0.1086**	0.0157**	0.0119***	-0.0461	-0.0017	0.0054
	(0.0415)	(0.0056)	(0.0034)	(0.1324)	(0.0157)	(0.0133)
Observations	1537	1537	1537	149	149	149
Adjusted R ²	0.032	0.056	0.045	0.028	0.175	0.080

Notes: This table reports the results of six OLS regressions estimating the effect of treatment on Black speaker representation, estimated separately for seminars with no Black organizer (Models 1-3; $N=1,537$) and seminars with at least one Black organizer (Models 4-6; $N=149$). All models include fixed effects for (1) the seminar's discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted), and also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URM in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses. In our sample, 89% of seminars had non-Black organizers so would, in theory, benefit from bridging connections to Black speakers. Exploratory analyses examined treatment effects among seminars with Black vs. non-Black organizers across three outcomes: the percentage of speakers who were Black, the number of Black

speakers, and the probability of inviting any Black speaker. Results are massively underpowered to compare treatment effects, but if anything suggest treatment effects were directionally stronger in seminars with non-Black organizers (% Black: $B = 0.74$, $SE = 0.32$, $p = 0.019$; # Black: $B = 0.08$, $SE = 0.04$, $p = 0.055$; Any Black: $B = 0.05$, $SE = 0.02$, $p = 0.032$) rather than Black organizers (% Black: $B = -0.57$, $SE = 1.35$, $p = 0.674$; # Black: $B = -0.08$, $SE = 0.18$, $p = 0.660$; Any Black: $B = -0.004$, $SE = 0.112$, $p = 0.975$). For analogous analyses of URM and Hispanic speaker outcomes by organizer race, see Extended Tables E13–E14 on OSF.

$\dagger p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Section 7: References

- Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In A. V. Banerjee & E. Duflo (Eds.), *Handbook of Economic Field Experiments* (Vol. 1, pp. 73–140). North-Holland.
- Hansen, B. B., & Bowers, J. (2008). Covariate balance in simple, stratified and clustered comparative studies. *Statistical Science*, 23(2), 219–236.
- Jaro, M. A. (1989). Advances in record-linkage methodology as applied to matching the 1985 Census of Tampa, Florida. *Journal of the American Statistical Association*, 84(406), 414–420.
- Imai, K., & Khanna, K. (2016). Improving ecological inference by predicting individual ethnicity from voter registration records. *Political Analysis*, 24(2), 263–272.
- Kozlowski, A. C., Taddy, M., & Ashmore, J. D. (2022). Avoiding bias when inferring race using name-based approaches. *PLOS ONE*, 17(3), e0264270.
- Mogstad, M., Romano, J. P., Shaikh, A., & Wilhelm, D. (2024). Inference for ranks with applications to mobility across neighbourhoods and academic achievement across countries. *Review of Economic Studies*, 91(1), 476–518.
- Parkhi, O. M., Vedaldi, A., & Zisserman, A. (2015). Deep face recognition. In X. Xie, M. W. Jones, & G. K. L. Sherwin (Eds.), *Proceedings of the British Machine Vision Conference* (pp. 41.1–41.12). BMVA Press.
- Winkler, W. E. (1990). String comparator metrics and enhanced decision rules in the Fellegi-Sunter model of record linkage. *Proceedings of the Section on Survey Research Methods*, American Statistical Association, 354–359.